

NAYAB NAYYER

AI/ML Engineer | Python | Machine Learning & Deep Learning
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[LinkedIn](#) | [Github](#)

PROFESSIONAL SUMMARY

Computer Science graduate and Silver Medalist with hands-on experience in Python, machine learning, NLP, and deep learning through an AI internship and practical projects. Experience building ML applications, recommendation systems, semantic-search/LLM applications, and CNN-based computer vision systems using Scikit-learn and TensorFlow. Currently strengthening AI/ML engineering skills through practical work in model development, evaluation, and deployment.

SKILLS

Programming & Data: Python, SQL, Pandas, NumPy

Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation, Classification, Regression

Deep Learning & Computer Vision: TensorFlow, Keras, CNNs, Neural Networks, Image Classification, Grad-CAM

NLP & Generative AI: LLM Applications, Gemini API, FAISS, Semantic Search, TF-IDF, Cosine Similarity, Text Preprocessing

Development & Deployment: Git, GitHub, Streamlit, Streamlit Cloud

Currently Learning: FastAPI, Docker, RAG, LangChain, MLOps

PROFESSIONAL EXPERIENCE

AI Intern

Evolvian Softwares | Islamabad, Pakistan | June 2026 – Present

- Completed structured hands-on ML training covering core concepts and algorithms through the CampusX curriculum.
- Built a movie recommendation system using KNN and cosine similarity.
- Used Git and GitHub for version control and project workflows.
- Currently strengthening TensorFlow and deep learning through practical implementation.

EDUCATION

Bachelor of Science in Computer Science

Lahore College for Women University | Lahore, Pakistan

CGPA: 3.64/4.00 | Silver Medalist (2nd Position) | June 2026

PROJECTS

Explainable Pneumonia Detection from X-rays – Ongoing Project

Python | TensorFlow | Keras | CNN | Grad-CAM | Image Classification

- Developing a CNN-based system for binary classification of chest X-ray images into NORMAL and PNEUMONIA classes.
- Designed and trained a custom CNN using TensorFlow/Keras with data preprocessing, model training, validation, and independent test evaluation.
- Evaluating model performance using accuracy, precision, recall, F1-score, and confusion matrix analysis, with current evaluation showing 97.95% recall for the PNEUMONIA class and 83.86% F1-score.

- Conducting error analysis and developing Grad-CAM-based explanations to investigate visual regions influencing predictions.

MedPredict Pro – Medical Cost Prediction Web Application

Python | Scikit-learn | Pandas | NumPy | Streamlit

- Developed and deployed an end-to-end machine learning web application for predicting medical insurance costs.
- Structured the codebase into clean, modular functions with error handling for invalid user inputs.
- Built and trained a Random Forest Regression model using Scikit-learn, achieving an R^2 score of 0.89.
- Performed data preprocessing, feature engineering, model evaluation and Streamlit deployment.

GitHub: [View Repository](#)

DecodeBot – AI Chatbot Assistant

Python | Streamlit | FAISS | Gemini API

- Built a hybrid AI chatbot using rule-based NLP, semantic search, and Gemini LLM integration.
- Developed a decision-routing system for intent recognition, semantic matching, and fallback responses.
- Integrated calculator, weather, and web search tools for enhanced user interaction.
- Designed a multi-chat Streamlit interface with session-based memory and conversation history.

GitHub: [View Repository](#)

PathAI – AI Career Recommendation System

Python | Scikit-learn | TF-IDF | Cosine Similarity | Streamlit | Plotly

- Built an AI-powered career recommender using TF-IDF vectorization and cosine similarity to match user skills and interests against real candidate profiles.
- Implemented keyword expansion and sublinear TF-IDF scaling to improve recommendation accuracy for short and varied inputs.
- Developed explainable AI recommendations with matched skills, skill gap analysis, and career growth roadmaps.
- Designed an interactive Streamlit dashboard with match score cards, radar chart, bar charts, and skill gap priority visualization.

GitHub: [View Repository](#)

CERTIFICATIONS

- Machine Learning Specialization — Coursera (Audit Track) | March 2025
- Python for Data Science — Kaggle | November 2024
- Introduction to Cybersecurity — HarvardX | June 2025

LANGUAGES

English (Fluent), Urdu (Native), Punjabi (Fluent)